

End – Semester Examination (Autumn Semester) 2023 – 24  
(For 3<sup>rd</sup> Year 5<sup>th</sup> Semester B.Tech)

AU/CSE/Odd/2023 – 24

Paper Name: Design and Analysis of Algorithms  
Paper Code: CSEUGPC12

Full Marks: 80  
Time: 3 Hrs

Section – A

Answer any five of the following  $5 \times 2 = 10$

1. A machine needs a minimum of 100 seconds to sort 1000 names by quick sort. What will be the approximate minimum time needed (in seconds) to sort 100 names?
2. Describe the characteristics of an algorithm.
3. State the recurrence relations for the binary search and quick sort algorithm.
4. A complete binary tree with 'n' non-leaf nodes contains \_\_\_ nodes.
5. Which among the insertion sort, quick sort, merge sort and heap sort perform in least time in the worst case?
6. The way a card game player arranges his cards as he picks them up one by one, is an example of \_\_\_\_\_ sort.
7. What are the best and worst space complexity of the quick sort algorithm?

Section – B

Answer any five of the following  $5 \times 6 = 30$

1. State the Big-Oh, Big-Omega notations. Explain their graphical interpretations.
2. Find the longest common subsequences between  $X = \{AGACGCG\}$  and  $Y = \{GAGCC\}$  using dynamic programming paradigm.
3. Explain the Prim's algorithm implemented through min-heap.
4. You have an array of size 15 in which 12 numbers are arranged into a max-heap. If you want to add a new number into the heap, how to do that? Write the pseudo-code for the above problem.
5. From the recurrence relation, obtain the time complexity of the merge sort algorithm through the back substitution method.
6. Explain the algorithm for matrix chain multiplication problem using dynamic programming approach.
7. Solve the recurrence relation using the recursion tree method:  $T(n) = 2T(n/2) + n^2$ .

Section – C

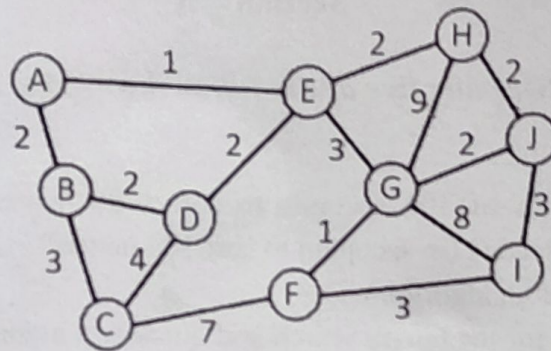
Answer any four of the following  $4 \times 10 = 40$

1. What is dynamic programming? Compare and contrast between dynamic programming and greedy method. Describe the dynamic programming algorithm to find the longest common subsequences (LCS) between two strings. [2 + 2 + 6]



2. Deduce the time complexity of the Strassen's matrix multiplication by solving the recurrence relation using the recursion tree method. The matrices  $A_1, A_2, A_3, A_4$  have the following dimensions  $10 \times 5, 5 \times 20, 20 \times 10$ , and  $10 \times 5$ , respectively. Explain the dynamic programming approach to find an optimal parenthesization of the above matrix chain product. [4 + 6]

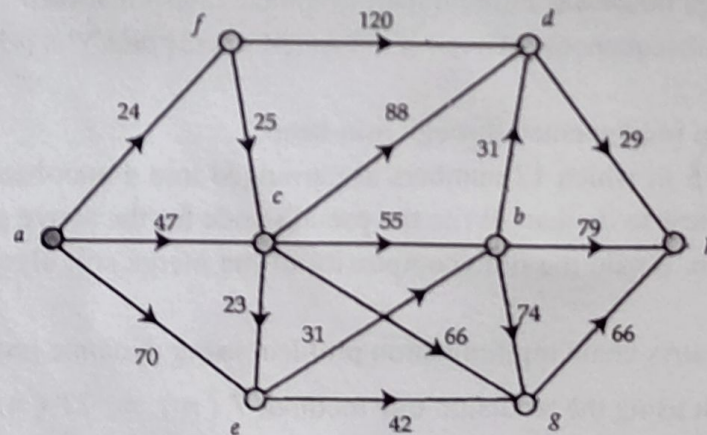
3. What is a spanning tree? How many spanning trees is possible from a graph with 7 nodes? What is a minimum cost spanning tree? Show the step-by-step procedure to find a minimum spanning tree of the given graph by the Kruskal algorithm. [2 + 1 + 1 + 6]



4. Find the percentage of improvement on variable length code over fixed length coding through Huffman encoding when the frequencies of characters a to f are as follows: a : 12, b : 15, c : 3, d : 30, e : 25, f : 15. Let {17, 10, 35, 12, 20, 2, 15, 30, 80, 1} be a list of 10 numbers. Provide a step-by-step explanation of how a heap sort algorithm can be used to sort the list. [5 + 5]

5. State the 0/1 Knapsack problem. Consider instance of the 0/1 knapsack problem as below with  $P$  depicting the profit value and  $W$  depicting the weight of each item whereas  $M$  denotes the total weight carrying capacity of the knapsack. Find optimal answer using dynamic programming technique.  $P: [40, 10, 50, 30, 60]$   $W: [80, 10, 40, 20, 90]$   $M: 110$ . If fractional part of items is allowed to pick then what will be the optimal solution? [2 + 5 + 3]

6. a) Find the shortest paths from node 'a' to all vertices using Dijkstra algorithm for the following graph.



- b) State the master theorem. Use the master method to find the tight asymptotic bounds for the following recurrences:

- $T(n) = 3T(n/2) + n \log n$
- $T(n) = 9T(n/3) + n$

[6 + 2 + 2]